Technology Stack

Date	15 March 2024
Team ID	UNM-293
Project Name	AI Comic Story Creator using Gemini Models
Maximum Marks	2 Marks

Define Your Technology Stack:
Identify programming languages, frameworks, databases, front-end, back-end, and APIs used.

Technology Stack Details

S.No	Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, HTML/CSS)	Streamlit + HTML/CSS + JavaScript + Pillow (PIL) for image handling	Responsive design, Rapid prototyping, Built-in widgets for comic preview, Easy integration with Python backend, Component reusability
2	Backend / Server-Side (e.g., Node.js, Python)	Python 3.9+ Flask (optional) Google Generative AI SDK gemini-1.5-flash	Fast processing, Excellent AI library support, Existing expertise in Python, Async handling for Gemini API, Easy deployment
3	Database / Data Storage (e.g., MySQL, MongoDB)	Streamlit Session State Local File Storage (/output) Future: Firebase / MongoDB	Handles temporary comic data, No complex relations need ed for MVP, Strong for file storage, Easy session management
4	Cloud / Hosting / Deployment (e.g., AWS, Heroku)	Streamlit Cloud GitHub Render (backup) Docker (optional)	Easy scalability, Free hosting for Streamlit apps, Cost -effective, CI/CD via GitHub push, One-click deploy
5	Version Control & CI/CD (e.g., GitHub, Jenkins)	Git + GitHub GitHub Actions Streamlit Cloud auto-deploy	Seamless collaboration (even solo), Automated testing, Version history, Backup of code, Automated deployment on push
6	Third-Party APIs / Other Tools (e.g., Stripe, Google Maps)	Google Gemini API - gemini-1.5-flash (story) - Imagen / Gemini Vision (images) + FPDF for PDF export	Enables core feature: AI story & image generation, High -quality generative AI, Free tier available, Powers entire comic creation pipeline